

Differential Equation

MT-221

Course Contents

- ① linear algebra
- ② 1st order Ordinary Differential Equation
- ③ 2nd order ODE's
- ④ Partial Differential Equation
- ⑤ Fourier Series

CLO-1: Describe formation of differential equations and system of linear equation to explain physical situations [PLO-1]

CLO-2: Apply appropriate methods to solve differential equations and systems of linear equations of relevant engineering problem [PLO-2]

Recommended Books:

- ① Elementary Linear Algebra by Howard Anton.
- ② Differential Equations by C. Zill (Practice Prob)
- ③ Advance Engineering Mathematics by Erwin Kreyszig (For theories)

Sessional Marks Criteria

20 marks for mid term

15 marks for assignments

5 marks for Quiz.

DIFFERENTIAL EQUATION

An equation which contain differential coefficient is called differential equation.

The differential equation are of two types:

- 1) Ordinary differential equation (ODEs)
- 2) Partial differential equation (PDEs)

$$y = f(x)$$

↳ input variable / Independent variable

↳ Output variable / Dependent variable / Range

On the basis of number of input variable, we give function its name.

$y = f(x) \rightarrow$ function of one variable.

$\frac{dy}{dx} \rightarrow$ dependent
 $dx \rightarrow$ independent

when we differentiate on multivariable then we partially differentiate it.

ORDINARY DIFFERENTIAL EQUATION:

An ODE is an equation that contain one or several derivatives of an unknown function which we usually called $y(x)$

Example:

$$i) \frac{dy}{dx} = \cos x \quad ; \quad \frac{d^2y}{dx^2} + 9y = e^{-2x}$$

$$\frac{dy}{dx} = \frac{1+x^2}{1-y^2} \quad ; \quad \frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 3y = 0 \quad (2nd \text{ order})$$

$$\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2} = k \frac{d^2y}{dx^2}$$

Order & Degree Of Differential Eq:

The order of the differential is the order of the highest differential coefficient present in the equation.

The highest power of highest derivative in a differential equation gives the degree of that equation and it must be made free from radical and fractions as derivatives are concerned.

Example:

$$\cos x \frac{d^2 y}{dx^2} + \sin x \left[\frac{dy}{dx} \right]^2 + 8y = \tan x$$

Order : 2 and Degree : 1

$$\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^3 = \left[\frac{d^2 y}{dx^2} \right]^2$$

Order : 2 and Degree : 2

FORMATION OF DIFFERENTIAL EQUATION:

A differential equation can be formed by differentiating the ordinary equation and eliminating the arbitrary constants

$$y = x^2 + C$$

→ From this we get family of curves
we get graphs of same shape.

Q: Form the differential equation by eliminating the arbitrary c , and also write down the degree and order of differential equation obtained.

$$i) \quad y = Ax + A^2$$

$$\frac{dy}{dx} = A$$

$$y = \frac{dy}{dx} x + \left(\frac{dy}{dx} \right)^2$$

Order = 1 ; degree = 2

ii) $y = A \cos x + B \sin x$

$$\frac{dy}{dx} = -A \sin x + B \cos x$$

$$\frac{d^2y}{dx^2} = -A \cos x - B \sin x = -(A \cos x + B \sin x)$$

$$\frac{d^2y}{dx^2} = -y$$

$$\frac{d^2y}{dx^2} + y = 0$$

Order : 2 and degree : 1

iii) $y^2 = Ax^2 + Bx + C$

$$2y \frac{dy}{dx} = 2Ax + B$$

$$\cancel{x} \left\{ y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 \right\} = \cancel{x} A$$

$$y \frac{d^3y}{dx^3} + \frac{d^2y}{dx^2} \cdot \frac{dy}{dx} + 2 \frac{dy}{dx} \cdot \frac{d^2y}{dx^2} = 0$$

$y \frac{d^3y}{dx^3} + 3 \frac{dy}{dx} \cdot \frac{d^2y}{dx^2} = 0$	Ans
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H.W

Q Form the differential equation

i) $y = ax + bx^2 + cx^3$

ii) $y = A \cos 2t + B \sin 2t$

iii) $y = ae^x + be^{2x} + ce^{3x}$

iv) $y = ae^x + be^{-x} + c \cos x + d \sin x$

Solution Of Differential Equation

Differential Equations Of Order One

And Degree One:

The standard methods of solving the differential equation of the following types are

- i) Equations solvable by separation of variables.
- ii) Homogenous differential equation.
- iii) Linear differential equation
- iv) Exact differential equation.

i) Variable Separable Method:

If a differential equation can be written in the form

$$f(y) dy = g(x) dx$$

Q Solve the given differential equation :-

$$i) (x+1) \frac{dy}{dx} = x(y^2+1)$$

Solution :

$$(x+1) dy = x(y^2+1) dx$$

$$\Rightarrow \frac{1}{(y^2+1)} dy = \frac{x}{(x+1)} dx$$

Integrate

$$\Rightarrow \frac{1}{1+y^2} dy = \frac{x+1-1}{(x+1)} dx$$

$$\Rightarrow \frac{1}{(1+y^2)} dy = \left\{ \frac{(x+1)}{(x+1)} - \frac{1}{(x+1)} \right\} dx$$

$$\Rightarrow \frac{1}{1+y^2} dy = \left(1 - \frac{1}{1+x} \right) dx$$

Integrate on both sides

$$\Rightarrow \int \frac{1}{(1+y^2)} dy = \int dx - \int \frac{1}{1+x} dx$$

$$\Rightarrow \tan^{-1}(y) = x - \ln(1+x) + C$$

$$\tan^{-1}y = x - \ln(1+x) + C$$

OR

$$y = \tan(x - \ln(1+x) + C)$$

Ans

$$\text{ii) } (xy^2 + x) dx + (x^2y + y) dy = 0$$

$$x(y^2 + 1) dx + y(x^2 + 1) dy = 0$$

$$y(x^2 + 1) dy = -x(y^2 + 1) dx$$

$$\left(\frac{y}{y^2 + 1} \right) dy = \frac{-x}{(x^2 + 1)} dx$$

Integrate on both sides

$$\frac{1}{2} \int \frac{1}{(y^2 + 1)} (2y) dy = -\frac{1}{2} \int \frac{1}{(x^2 + 1)} (2x) dx$$

$$\frac{1}{2} \ln(y^2 + 1) = -\frac{1}{2} \ln(x^2 + 1) + C$$

$$\frac{1}{2} \ln(y^2 + 1) + \frac{1}{2} \ln(x^2 + 1) = C$$

$$\frac{1}{2} \ln \{(y^2 + 1)(x^2 + 1)\} = C$$

$$\ln \sqrt{(y^2 + 1)(x^2 + 1)} = C$$

$$\ln |y^2 + 1| = -\ln |x^2 + 1| + C$$

$$\ln(y^2 + 1) + \ln(x^2 + 1) = C$$

$$\ln(y^2 + 1)(x^2 + 1) = C$$

Taking exponential on b/s

$$(y^2 + 1)(x^2 + 1) = e^C$$

$$(y^2 + 1)(x^2 + 1) = C$$

Date: _____

Q) $x \frac{dy}{dx} + \cot y = 0$, initial condition $y = \frac{\pi}{4}$, $x = \sqrt{2}$

$$x \frac{dy}{dx} = -\cot y$$

$$\tan y \, dy = -\frac{1}{x} \, dx$$

Integrate on b/s

$$\int \tan y \, dy = -\int \frac{1}{x} \, dx$$

$$\ln |\sec y| = -\ln |x| + C$$

$$\ln |\sec y| = -\ln |x| + \ln C$$

$$\ln |\sec y| = \ln \left(\frac{C}{x} \right) \quad \text{OR Applying exp on both sides}$$

$$\sec y = \frac{C}{x}$$

$$\frac{x}{\cos y} = C$$

$$\boxed{C = 2} \quad \text{Putting}$$

$$\frac{x}{\cos y} = 2$$

$$\frac{x}{2} = \cos y$$

$$\boxed{y = \cos^{-1} \left(\frac{x}{2} \right)}$$

Date: _____

$$\text{iv) } x^4 \frac{dy}{dx} + x^3 y = -\sec(xy)$$

By substitution method

let

$$t = xy$$

$$\frac{dt}{dx} = x \frac{dy}{dx} + y$$

$$x^4 \frac{dy}{dx} + x^3 y = -\sec(xy)$$

$$x^3 \left(x \frac{dy}{dx} + y \right) = -\sec(xy)$$

$$x^3 \left(\frac{dt}{dx} \right) = -\sec t$$

$$\cos t \, dt = -\frac{1}{x^3} dx$$

$$\int \cos t \, dt = -\int x^{-3} dx$$

$$\sin t = -\left(\frac{x^{-2}}{-2} \right) + C$$

$$\sin t = \frac{1}{2x^2} + C$$

$$\boxed{\sin(xy) - \frac{1}{2x^2} = C} \quad \text{Ans}$$

Homogenous Differential Equation:

A differential equation of the form

$$\frac{dy}{dx} = \frac{f(x, y)}{g(x, y)} \rightarrow (1)$$

$$= F(x, y)$$

is homogenous iff

$$F(\lambda x, \lambda y) = \lambda^n F(x, y)$$

E.g

Check whether the eq is homogenous or not

i) $F(x, y) = \frac{2x^2y}{x^2 + y^2}$

$$F(\lambda x, \lambda y) = \frac{2(\lambda x)^2(\lambda y)}{(\lambda x)^2 + (\lambda y)^2} = \frac{\lambda^3 (2x^2y)}{\lambda^2 (x^2 + y^2)}$$

$$F(\lambda x, \lambda y) = \lambda F(x, y)$$

The homogenous is
equation is

ii) $F(x, y) = \frac{2xy}{x^2 + y^2}$

$$F(\lambda x, \lambda y) = \frac{2(\lambda x)(\lambda y)}{(\lambda x)^2 + (\lambda y)^2} = \frac{\lambda^2 (2xy)}{\lambda^2 (x^2 + y^2)}$$

$$F(\lambda x, \lambda y) = \lambda^0 \frac{(2xy)}{(x^2 + y^2)} = \lambda^0 F(x, y)$$

Homogenous Equation.

Date: _____

$$\text{iii) } F(x, y) = \ln(x) - \ln(y) + \frac{x+y}{x-y}$$

$$F(\lambda x, \lambda y) = \ln(\lambda x) - \ln(\lambda y) + \frac{\lambda x + \lambda y}{\lambda x - \lambda y}$$

$$F(\lambda x, \lambda y) = \ln\left(\frac{\lambda x}{\lambda y}\right) + \frac{\lambda(x+y)}{\lambda(x-y)}$$

$$F(\lambda x, \lambda y) = \ln\left(\frac{x}{y}\right) + \frac{(x+y)}{(x-y)}$$

$$F(\lambda x, \lambda y) = \ln(x) - \ln(y) + \frac{(x+y)}{(x-y)}$$

The equation is homogenous.

$$\text{iv) } \frac{dy}{dx} = \frac{y}{x} + x \sin y$$

Here

$$F(x, y) = \frac{y}{x} + x \sin y$$

$$F(\lambda x, \lambda y) = \frac{\lambda y}{\lambda x} + \lambda x \sin\left(\frac{\lambda y}{\lambda x}\right)$$

$$F(\lambda x, \lambda y) = \frac{y}{x} + \lambda x \sin\left(\frac{y}{x}\right)$$

Non-homogenous Equation

$$F(\lambda x, \lambda y) \neq F(x, y)$$

Solution Of Homogenous Differential

Equation :

$$y = vx$$
$$\frac{dy}{dx} = V + x \frac{dv}{dx}$$

$$\therefore \frac{dy}{dx} = \frac{f(x,y)}{g(x,y)}$$

- This substitution is valid for all homogenous differential equation
- The reduced equation involved only V and x .
- This new differential equation can be solved by variable separable method.

Q1 Solve the given differential equation

$$(2xy + x^2) y' = 3x^2 + 2xy$$

$$y' = \frac{3x^2 + 2xy}{(2xy + x^2)}$$

$$\frac{dy}{dx} = \frac{3x^2 + 2xy}{2xy + x^2} \rightarrow (1)$$

Here

$$F(x,y) = \frac{3x^2 + 2xy}{2xy + x^2}$$

$$F(\lambda x, \lambda y) = \frac{3(\lambda x)^2 + 2(\lambda x)(\lambda y)}{2(\lambda x)(\lambda y) + (\lambda x)^2}$$

Date: _____

$$F(\lambda x, \lambda y) = \frac{3\lambda^2 y^2 + 2\lambda^2 xy}{2\lambda^2 xy + \lambda^2 x^2}$$

$$F(\lambda x, \lambda y) = \frac{\cancel{\lambda^2} (3y^2 + 2xy)}{\cancel{\lambda^2} (2xy + x^2)}$$

$$F(\lambda x, \lambda y) = \lambda^0 F(x, y)$$

The eq. is homogenous

Put $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{3(vx)^2 + 2x(vx)}{2x(vx) + (x)^2} \quad (\text{From eq. (1)})$$

$$v + x \frac{dv}{dx} = \frac{3v^2 x^2 + 2vx^2}{2vx^2 + x^2}$$

$$v + x \frac{dv}{dx} = \frac{\cancel{x^2} (3v^2 + 2v)}{\cancel{x^2} (2v + 1)}$$

$$v + x \frac{dv}{dx} = \frac{3v^2 + 2v}{2v + 1}$$

$$x \frac{dv}{dx} = \frac{3v^2 + 2v}{2v + 1} - v$$

$$x \frac{dv}{dx} = \frac{3v^2 + 2v - 2v^2 - v}{2v + 1}$$

$$\frac{2v+1}{v^2+v} dv = \frac{1}{x} dx$$

$$\ln(v^2 + v) = \ln(x) + C$$

$$\ln(v^2 + v) = \ln(xC)$$

Apply exp on b/s

$$v^2 + v = xC$$

$$\text{Put } v = \frac{y}{x}$$

$$\frac{y^2}{x^2} + \frac{y}{x} = xC$$

$$y^2 + xy = x^3 C$$

Q2 Solve the given differential equation:

$$x \sin \frac{y}{x} dy = \left[y \sin \frac{y}{x} - x \right] dx$$

Solution :

$$\frac{dy}{dx} = \frac{y \sin \left(\frac{y}{x} \right) - x}{x \sin \left(\frac{y}{x} \right)}$$

$$F(x, y) = \frac{y \sin \left(\frac{y}{x} \right) - x}{x \sin \left(\frac{y}{x} \right)}$$

$$F(\lambda x, \lambda y) = \frac{\lambda y \sin \left(\frac{\lambda y}{\lambda x} \right) - \lambda x}{\lambda x \sin \left(\frac{\lambda y}{\lambda x} \right)}$$

$$F(\lambda x, \lambda y) = \frac{y \sin \left(\frac{y}{x} \right) - x}{x \sin \left(\frac{y}{x} \right)}$$

$$\text{Put } y = vx$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{(vx) \sin\left(\frac{vx}{x}\right) - x}{x \sin\left(\frac{vx}{x}\right)}$$

$$v + x \frac{dv}{dx} = \frac{\cancel{x} \{v \sin(v) - 1\}}{x \sin(v)}$$

$$x \frac{dv}{dx} = \frac{v \sin v - 1 - v}{\sin v}$$

$$x \frac{dv}{dx} = \frac{\cancel{v \sin v} - 1 - \cancel{v \sin v}}{\sin v}$$

$$x \frac{dv}{dx} = \frac{-1}{\sin v}$$

$$\sin v \, dv = \frac{-1}{x} dx$$

$$\int \sin v \, dv = - \int \frac{1}{x} dx$$

$$\cancel{\cos v} \, dv =$$

$$-\cos v = -\ln(x) + C$$

$$-\cos \frac{y}{x} + \ln(x) = C$$

$$\boxed{\cos \frac{y}{x} - \ln(x) = C}$$

Equation Reducible to Homogenous

Form, =

The equations of the form.

$$\frac{dy}{dx} = \frac{ax + by + c}{Ax + By + C}$$

can be reduced to homogenous form by ^{using suitable} substitution

Case I:

If the ratios are not same

$$\text{i.e. } \frac{a}{A} \neq \frac{b}{B}$$

use $x = X + h$ and $y = Y + k$

$$\frac{dy}{dx} = \frac{dY}{dX}$$

The given differential equation reduces to

$$\frac{dY}{dX} = \frac{a(X+h) + b(Y+k) + C}{A(X+h) + B(Y+k) + C}$$

$$= \frac{aX + bY + ah + bk + C}{AX + BY + Ah + Bk + C}$$

$$= \frac{aX + bY + ah + bk + C}{AX + BY + Ah + Bk + C}$$

choose h, k so that $ah + bk + C = 0$

$$Ah + Bk + C = 0$$

Then the given equation becomes

$$\frac{dY}{dX} = \frac{aX + bY}{AX + BY}$$

This eq is homogenous

Case II

If ratios are same

$$\frac{a}{A} = \frac{b}{B} = \frac{1}{m} \quad (\text{Say})$$

The given equation becomes

$$\frac{dy}{dx} = \frac{ax+by+c}{m(ax+by)+C}$$

Now put $z = ax + by$ and apply variable separable method.

Ex: 1

$$(x+2y)(dx-dy) = dx+dy$$

$$x dx + 2y dx - x dy - 2y dy = dx + dy$$

$$x dx + 2y dx - x dy - 2y dy - dx - dy = 0$$

$$x dx + 2y dx - dx - x dy - 2y dy - dy = 0$$

$$dx(x+2y-1) - dy(x+2y+1) = 0$$

$$\frac{dy}{dx} = \frac{(x+2y-1)}{(x+2y+1)}$$

The equation is not separable and not by homogenous because constant exist.

$$\therefore \frac{dy}{dx} = \frac{Ax+by+c}{Ax+By+C}$$

$$\frac{a}{A} = \frac{b}{B} \quad \text{or} \quad \frac{1}{1} = \frac{2}{2} \quad \text{or} \quad \frac{1}{1} = \frac{1}{1}$$

The ratios are same.

let

$$z = x + 2y$$

$$\frac{dz}{dx} = 1 + 2 \frac{dy}{dx}$$

$$\frac{1}{2} \left[\frac{dz}{dx} - 1 \right] = \frac{dy}{dx}$$

$$\frac{1}{2} \left\{ \frac{dz}{dx} - 1 \right\} = \frac{z-1}{z+1}$$

$$\frac{dz}{dx} = \frac{2z-2}{z+1} + 1 = \frac{2z-2+z+1}{z+1}$$

$$\frac{dz}{dx} = \frac{3z-1}{z+1}$$

$$\frac{z+1}{3z-1} dz = dx$$

$$\int \frac{z+1}{3z-1} dz = \int dx$$

$$\int \frac{(3z-2z+3-2)}{3z-1} dz = \int dx$$

$$3(z+1) - 2(z+1)$$

$$\int \left(\frac{z}{3z-1} + \frac{1}{3z-1} \right) dz = \int dx$$

$$\frac{1}{3} \ln |3z-1| +$$

$$\int \frac{z+1}{3z-1} dz = \int dx$$

By applying partial fraction

$$\frac{z+1}{3(z-\frac{1}{3})} = \frac{1}{3} + \frac{A}{(z-\frac{1}{3})}$$

$$\frac{z+1}{3(z-\frac{1}{3})} = \frac{(z-\frac{1}{3}) + 3A}{3(z-\frac{1}{3})}$$

$$z+1 = (z-\frac{1}{3}) + 3A$$

$$\text{Let } z = \frac{1}{3}$$

$$\frac{1}{3} + 1 = \frac{1}{3} - \frac{1}{3} + 3A$$

$$\boxed{\frac{4}{9} = A}$$

$$\frac{z+1}{3(z-\frac{1}{3})} = \frac{1}{3} + \frac{1}{9} \times \frac{4}{(z-\frac{1}{3})}$$

$$\frac{z+1}{3(z-\frac{1}{3})} = \frac{1}{3} + \frac{1}{9} \times \frac{4 \times 3}{3z-1}$$

$$\frac{z+1}{3(z-\frac{1}{3})} = \frac{1}{3} \left(1 + \frac{4}{3z-1} \right)$$

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$$\frac{1}{3} \int \left(1 + \frac{4}{3z-1} \right) dz = \int dx$$

$$\frac{1}{3} \left[\int dz + 4 \int \frac{1}{3z-1} dz \right] = x + C$$

$$\frac{1}{3} \left[z + \frac{4}{3} \ln |3z-1| \right] = x + C$$

$$\frac{1}{3} z + \frac{4}{9} \ln |3z-1|$$

$$\frac{1}{3} (x+2y) + \frac{4}{9} \ln |3(x+2y)-1| = x + C$$

$$\frac{(x+2y)}{3} + \frac{4}{9} \ln |3x+6y-1| = x + C$$

$$3x+6y + 4 \ln |3x+6y-1| = 9x+9C$$

$$3x - 9x + 6y + 4 \ln |3x+6y-1| = C$$

$$-6x + 6y + 4 \ln |3x+6y-1| = C$$

$$-2(3x-3y - 2 \ln |3x+6y-1|) = C$$

$$\cancel{3x+3y}$$

$$3x - 3y + \frac{1}{2}C = 2 \ln |3x+6y-1|$$

$$3x - 3y + C = 2 \ln |3x+6y-1|$$

Ans

Ex no 2

$$\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$$

Solution

$$\frac{dy}{dx} = \frac{ax+by+c}{a_1x+b_1y+c_1}$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} ; \quad \frac{1}{2} \neq \frac{2}{1}$$

$$\text{let } x = X+h \text{ and } y = Y+k$$

$$\frac{dY}{dX} = \frac{(X+h)+2(Y+k)-3}{2(X+h)+(Y+k)-3}$$

$$\frac{dY}{dX} = \frac{X+h+2Y+2k-3}{2X+2h+Y+k-3}$$

$$\frac{dY}{dX} = \frac{X+2Y+h+2k-3}{2X+Y+2h+k-3}$$

let

$$h+2k-3=0$$

$$2h+k-3=0$$

$$h+2k=3 \rightarrow (1)$$

$$2h+k=3 \rightarrow (2)$$

Multiply eq (1) by 2 and subtract from (2)

$$\begin{array}{r} 2h+4k = 6 \\ + 2h+k = 3 \\ \hline - \quad - \quad - \end{array}$$

$$3k = 3$$

$$\boxed{k=1}$$

$$h = 3 - 2k$$

$$h = 3 - 2$$

$$\boxed{h=1}$$

$$\frac{dy}{dx} = \frac{x+2y}{2x+y} \rightarrow (3)$$

let

$$y = v x$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = \frac{x + 2vx}{2x + vx}$$

$$v + x \frac{dv}{dx} = \frac{\cancel{x}(1+2v)}{\cancel{x}(2+v)}$$

$$x \frac{dv}{dx} = \frac{1+2v}{2+v} - v$$

$$x \frac{dv}{dx} = \frac{1 + \cancel{2v} - \cancel{v} - v^2}{2+v}$$

$$x \frac{dv}{dx} = \frac{1 - v^2}{2+v}$$

$$\frac{2+v}{1-v^2} dv = \frac{1}{x} dx$$

$$\frac{v+2}{(1-v^2)} dv = \frac{1}{x} dx$$

$$\left(\frac{v}{1-v^2} + \frac{2}{1-v^2} \right) dv = \frac{1}{x} dx$$

$$\int \frac{v}{1-v^2} dv + 2 \int \frac{1}{1-v^2} dv = \int \frac{1}{x} dx$$

$$-\frac{1}{2} \int \frac{-2v}{1-v^2} dv + 2 \left[\frac{1}{2} \ln \left| \frac{1+v}{1-v} \right| \right] = \ln|x| + c$$

$$-\frac{1}{2} \ln |1-v^2| + \ln \left| \frac{1+v}{1-v} \right| = \ln |X| + C$$

$$\ln \left| \frac{1}{\sqrt{1-v^2}} \right| + \ln \left| \frac{1+v}{1-v} \right| = \ln |X| + \ln C$$

$$\ln \left| \left(\frac{1+v}{1-v} \right) \cdot \frac{1}{\sqrt{1+v} \sqrt{1-v}} \right| = \ln (XC)$$

Taking exponential on b/s

$$\left\{ \frac{(1+v)^{1-\frac{1}{2}}}{(1-v)^{1+\frac{1}{2}}} \right\} = XC$$

$$(1+v)^{\frac{1}{2}} = XC$$

$$(1-v)^{3/2}$$

$$\frac{(1+v)^{\frac{1}{2} \times X}}{(1-v)^{3/2 \times X}} = X^2 C^2 \quad (\text{Taking square})$$

$$\frac{(1+v)}{(1-v)^3} = X^2 C^2$$

$$(1-v)^3$$

$$\left(1 + \frac{y}{x}\right) = X^2 C^2$$

$$\left(1 - \frac{y}{x}\right)^3$$

$$\frac{x+y}{x}$$

$$(x-y)^3$$

$$x^3$$

$$\frac{(x+y)}{(x-y)^3} = X^2 C^2$$

$$(x-y)^3$$

$$x^2$$

$$\frac{x^2 (x+y)}{(x-y)^3} = x^2 C^2$$

$$\frac{x-h + y-k}{(x-h - y+k)^3} = C^2$$

$$\frac{x-1 + y-1}{(x-1 - y+1)^3} = C^2$$

$$\boxed{\frac{x+y-2}{(x-y)^3} = C^2}$$

LINEAR DIFFERENTIAL EQUATIONS

A first order differential equations in the dependent variable 'y' and independent variable 'x' is said to be linear if it is or it can be written in the form.

$$\frac{dy}{dx} + P(x)y = Q(x) \rightarrow (i)$$

$$\text{OR } \frac{dx}{dy} + P(y)x = Q(y) \rightarrow (ii)$$

WORKING RULES:

1) Write the differential equation in standard form
{ i.e eq (i) }

Find the int. + . I. t. T.F = $e^{\int P(x) dx}$

$$\frac{x^2 (x+y)}{(x-y)^3} = x^2 e^1$$

$$\frac{x-h + y-k}{(x-h - y+k)^3} = e^1$$

$$\frac{x-1 + y-1}{(x-1 - y+1)^3} = e^2$$

$$\boxed{\frac{x+y-2}{(x-y)^3} = e^2}$$

LINEAR DIFFERENTIAL EQUATIONS

A first order differential equations in the dependent variable 'y' and independent variable 'x' is said to be linear if it is or it can be written in the form.

$$\frac{dy}{dx} + P(x)y = Q(x) \rightarrow (i)$$

$$\text{OR } \frac{dx}{dy} + P(y)x = Q(y) \rightarrow (ii)$$

WORKING RULES:

1) Write the differential equation in standard form
 { i.e eq (i) }

2) Find the integrating factor I.F = $e^{\int P(x) dx}$

③ Multiply eq. ① by I.F

$$I.F \times \frac{dy}{dx} + I.F \cdot P(x)y = I.F \times Q(x)$$

$$④ \frac{d}{dx} \left[\begin{array}{c} I.F \times \text{Dependent} \\ \text{Variable} \end{array} \right] = I.F \times Q(x)$$

$$⑤ d[I.F \times D.V] = I.F \times Q(x) dx$$

⑥ Integrating both sides.

Q. 1

$$(x-1)^3 \frac{dy}{dx} + 4(x-1)^2 y = (x+1) \rightarrow (i)$$

The dependent variable (y) has power 1 then it is linear.

Multiplying eq. ① by $\frac{1}{(x-1)^3}$

$$\frac{dy}{dx} + \frac{4y}{(x-1)} = \frac{(x+1)}{(x-1)^3} \rightarrow (2) \quad \text{This equation is linear in 'y'}$$

$$P(x) = \frac{4}{(x-1)}$$

$$I.F = e^{\int P(x)} = e^{\int \frac{4}{x-1} dx}$$

$$I.F = e^{4 \ln(x-1)}$$

$$I.F = e^{4 \ln(x-1)^4}$$

$$I.F = (x-1)^4$$

Multiply eq. (i) by IF

$$(x-1)^4 \frac{dy}{dx} + \frac{4y}{(x-1)} (x-1)^4 = \frac{(x+1)}{(x-1)^3} \times (x-1)^4$$

$$(x-1)^4 \frac{dy}{dx} + 4y (x-1)^3 = (x+1)(x-1)$$

$$\frac{d}{dx} \left[(x-1)^4 \times y \right] = (x^2 - 1)$$

$$d \left[(x-1)^4 \times y \right] = (x^2 - 1) dx$$

Integrate on b/s

$$\int d \left[(x-1)^4 \times y \right] = \int x^2 dx - \int dx$$

$$\int d \left\{ (x-1)^4 \times y \right\} = \frac{x^3}{3} - x + C$$

$$\boxed{\left\{ (x-1)^4 \times y \right\} = \frac{x^3}{3} - x + C}$$

Date: _____

$$\text{Ex 2} \quad \frac{dy}{dx} + \tan x \cdot y = \cos^2 x$$

$$\text{Integrating factor} = e^{\int \tan x \, dx}$$

$$I.F. = e^{\ln(\sec x)}$$

$$\boxed{I.F. = \sec x}$$

Multiplying it on b/s

$$(\sec x) \frac{dy}{dx} + (\sec x) \tan x \cdot y = (\cos^2 x) (\sec x)$$

$$(\sec x) \frac{dy}{dx} + (\sec x) (\tan x) y = \cos x$$

$$\frac{d}{dx} \left[(\sec x) \cdot y \right] = \cos x$$

$$d \{ (\sec x) \cdot y \} = \cos x \, dx$$

$$\int d \{ \sec x \cdot y \} = \int \cos x \, dx$$

$$(\sec x \cdot y) = + \sin x + C$$

$$\boxed{\left(\frac{y}{\cos x} \right) = \sin x + C} \quad \text{Ans}$$

$$\text{Ex 3} \quad (x + 2y^3) \frac{dy}{dx} = y$$

$$\frac{dy}{dx} = \frac{y}{(x + 2y^3)}$$

$$\frac{dx}{dy} = \frac{(x + 2y^3)}{y}$$

$$\frac{dx}{dy} = \frac{x}{y} + 2y^2$$

$$\frac{dx}{dy} - \frac{1}{y} x = 2y^2 \rightarrow (1)$$

$$\therefore \frac{dx}{dy} + P(y)x = Q(y)$$

This equation is linear in 'x'

$$I.F. = e^{\int -\frac{1}{y} dy} = e^{-\ln(y)} = e^{\ln(\frac{1}{y})}$$

$$\boxed{I.F. = \frac{1}{y}}$$

Multiplying I.F. by eq. (1)

$$\frac{1}{y} \frac{dx}{dy} - \frac{1}{y^2} x = 2y$$

$$\frac{d}{dy} \left[\frac{1}{y} \cdot x \right] = 2y$$

$$\boxed{x - y^3 - y^2 C = 0}$$

$$d \left\{ \frac{1}{y} \cdot x \right\} = 2y dy$$

$$\int d \left\{ \frac{1}{y} \cdot x \right\} = \int 2y dy$$

$$\left(\frac{x}{y} \right) = y^2 + C$$

Equation Reducible To Linear Form: (Bernoulli's Equation)

An equation of the form:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \rightarrow (1)$$

is known as Bernoulli's equation. or

This equation is linear if $n=0$ & $n=1$, if $n \neq 0$ or $n \neq 1$ then by making suitable substitution Bernoulli's equation can be reduced in linear form.

Dividing eq. (1) by ' y^n '

$$(1) \Rightarrow \frac{1}{y^n} \frac{dy}{dx} + \frac{1}{y^{n-1}} P(x) = Q(x) \rightarrow (2)$$

Put $\frac{1}{y^{n-1}} = z \rightarrow$ Taking differentials

$$(1-n) \left(\frac{1}{y^n} \right) \frac{dy}{dx} = \frac{dz}{dx}$$

OR

$$\left(\frac{1}{y^n} \right) \frac{dy}{dx} = \frac{1}{(1-n)} \frac{dz}{dx}$$

$$2) \Rightarrow \frac{1}{(1-n)} \frac{dz}{dx} + P(x)z = Q(x)$$

$$\text{OR} \quad \frac{dz}{dx} + P(x)z = (1-n)Q(x)$$

This equation is linear and can be solved by the method of linear equation.

You have to suppose z , when the linear form of eq. ① changes.

Ex 1

$$\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2} \rightarrow \text{①}$$

This eq. is bernoulli because of y 's power of 2.

Divide eq. ① by y^2

$$\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{xy} = \frac{1}{x^2}$$

$$\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{x} \cdot \frac{1}{y} = \frac{1}{x^2} \rightarrow \text{②}$$

let suppose $z = \frac{1}{y}$

$$\frac{dz}{dx} = \frac{d}{dx} y^{-1}$$

$$\frac{dz}{dx} = -\frac{1}{y^2} \frac{dy}{dx}$$

$$\frac{1}{y^2} \frac{dy}{dx} = -\frac{dz}{dx} \quad \text{Put in eq. ②}$$

$$-\frac{dz}{dx} + \frac{1}{x} \cdot \frac{1}{y} = \frac{1}{x^2}$$

$$-\frac{dz}{dx} + \frac{1}{x} z = \frac{1}{x^2}$$

$$\frac{dz}{dx} - \frac{1}{x} z = -\frac{1}{x^2} \rightarrow \text{③}$$

The equation is linear in 'z' here $P(x) = -\frac{1}{x}$

$$I.F = e^{\int p(x) dx} = e^{\int -\frac{1}{x} dx} = e^{-\ln(x)} = \frac{1}{x}$$

$$\boxed{I.F = \frac{1}{x}}$$

Multiply eq (3) by I.F

$$\left(\frac{1}{x}\right) \frac{dz}{dx} - \frac{1}{x^2} z = -\frac{1}{x^3}$$

$$\frac{d}{dx} \left(\frac{1}{x} \times z \right) = -\frac{1}{x^3}$$

$$\int d \left(\frac{1}{x} \times z \right) = \int -x^{-3} dx$$

$$\left(\frac{1}{x} \times z \right) = - \left(\frac{x^{-2}}{-2} \right) + C$$

$$\frac{z}{x} = \frac{1}{2x^2} + C$$

$$\text{Put } z = \frac{1}{y}$$

$$\frac{1}{xy} = \frac{1}{2x^2} + C$$

$$\frac{1}{xy} - \frac{1}{2x^2} = C$$

~~$$xy = 2x^2 + \frac{1}{C}$$~~

$$\frac{1}{x} \left(\frac{1}{y} - \frac{1}{2x} \right) = C$$

~~$$\boxed{xy = 2x^2 + C}$$~~

$$\boxed{\left(\frac{1}{y} - \frac{1}{2x} \right) = xC}$$

Ex:

$$\frac{dy}{dx} + 4xy + xy^3 = 0$$

$$\frac{dy}{dx} + 4xy = -xy^3 \rightarrow (1)$$

Divide eq (1) by y^3

$$\frac{1}{y^3} \frac{dy}{dx} + \frac{4x}{y^2} = -x \Rightarrow \frac{1}{y^3} \frac{dy}{dx} + 4x \frac{1}{y^2} = -x$$

$$\text{let } z = \frac{1}{y^2}$$

$$\frac{dz}{dy} = -2y^{-3} \frac{dy}{dx} = -\frac{2}{y^3} \frac{dy}{dx}$$

$$\frac{2}{y^3} \frac{dy}{dx} = -\frac{dz}{dx}$$

$$\frac{1}{y^3} \frac{dy}{dx} = -\frac{1}{2} \frac{dz}{dx}$$

$$-\frac{1}{2} \frac{dz}{dx} + 4zx = -x$$

$$-\frac{dz/dx + 8zx}{2} = -x$$

$$-\frac{dz}{dx} + 8zx = -2x$$

$$\frac{dz}{dx} - 8zx = 2x \rightarrow (2) \quad \text{multiply eq by } -1$$

$$\text{I.F} = e^{\int -8x dx} = e^{-8 \frac{x^2}{2}} = \boxed{e^{-4x^2}}$$

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$$(e^{-4x^2}) \frac{dz}{dx} - 8x(e^{-4x^2})z = 2x(e^{-4x^2})$$

$$\frac{1}{e^{4x^2}} \frac{dz}{dx} - \frac{8x}{e^{4x^2}} z = \frac{2x}{e^{4x^2}}$$

$$\frac{d}{dx} \left(\frac{1}{e^{4x^2}} x z \right) = \frac{2x}{e^{4x^2}}$$

$$\int d \left(\frac{1}{e^{4x^2}} x z \right) = \int \frac{2x}{e^{4x^2}} dx$$

$$\left(\frac{1}{e^{4x^2}} x z \right) = \frac{1}{4} e^{4x^2} + C$$

$$\left(\frac{1}{e^{4x^2}} x z \right) = -\frac{1}{4} \frac{1}{e^{4x^2}} + C$$

$$\frac{z}{e^{4x^2}} + \frac{1}{4e^{4x^2}} = C$$

$$\frac{1}{e^{4x^2}} \left\{ z + \frac{1}{4} \right\} = C$$

$$\frac{1}{e^{4x^2}} \left\{ \frac{1}{y^2} + \frac{1}{4} \right\} = C$$

$$e^{-4x^2} \left\{ \frac{4+y^2}{4y^2} \right\} = C$$

Ans

Orthogonal Trajectories Of A Given Family Of Curves:

The general solution of first order differential eq contain one arbitrary constant and we have integral for each arbitrary constant. All these curves form a one parameter family of curve. We want to find out another family of curve which is orthogonal to the first family of curve.

$$m_1 \cdot m_2 = -1$$

Ex Find the family of orthogonal trajectories to the family
 $y = (x-k)^2$,

$$y = (x-k)^2 \rightarrow (1)$$

To remove k , take derivative

$$\frac{dy}{dx} = 2(x-k) \rightarrow (2)$$

Taking root on eq (1)

$$\sqrt{y} = (x-k) \quad (\text{Put in eq (2)})$$

$$\frac{dy}{dx} = 2\sqrt{y} \quad (m_1)$$

For the slope of the orthogonal trajectory $m_2 = -\frac{1}{m_1}$

$$\frac{dx}{dy} = -\frac{1}{2\sqrt{y}}$$

$$-dx = 2\sqrt{y} dy$$

$$-dx = 2y^{\frac{1}{2}} dy$$

Integrate

$$-\int dx = 2 \int y^{\frac{1}{2}} dy$$

$$-x = \frac{2 y^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$-x = \frac{4}{3} y^{\frac{3}{2}} + C$$

$$-\frac{3}{4}x = y^{3/2} + C$$

$$-\frac{3}{4}x$$

$$y = \left[-\frac{3}{4}(x+C) \right]^{2/3}$$

this is the required eq of orthogonal trajectory of the given curve.

$$x^2 - y^2 = C$$

$$2x - 2y \frac{dy}{dx} = 0$$

$$x \frac{dy}{dx} = y$$

$$y \frac{dy}{dx} = x \Rightarrow \frac{dy}{dx} = \frac{x}{y}$$

$$y \frac{dy}{dx} = x$$

∫

For the slope of orthogonal trajectory $m_2 = -\frac{1}{m_1}$

$$-\frac{dx}{dy} = \frac{x}{y}$$

$$\int -\frac{1}{x} dx = \int \frac{1}{y} dy$$

$$-\ln x = \ln y + C$$

$$-\ln x - \ln y = C$$

$$-\{\ln(xy)\} = C$$

$$\ln(xy) = -C$$

$$\ln(xy) = \ln C$$

$$\boxed{xy = C}$$



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Ex 4 $(D^4 + 6D^3 + 5D^2 - 24D - 36)y = 0$

The characteristic equation is

$$D^4 + 6D^3 + 5D^2 - 24D - 36 = 0$$

By using synthetic division method

2	1	6	5	-24	-36
		2	16	42	36
	1	8	21	18	0

Remainder

$$D^3 + 8D^2 + 21D + 18 = 0$$

Another Root is -2

-2	1	8	21	18
		-2	-12	-18
	1	6	9	0

Remainder

$$D^2 + 6D + 9 = 0$$

$$(D+3)^2 = 0$$

$$D = -3, -3$$

and

$$D = -3, -3, 2, -2$$

The complementary solution / Function is

$$y_c = (C_1 + C_2 x)e^{-3x} + C_3 e^{-2x} + C_4 e^{2x}$$

Ans

General Solution Of Higher Order Linear Ordinary

Differential Equation :- (Non Homogenous Equation)

General Solution = Complementary function +
Particular Integral

$$y = y_c + y_p$$

* Methods Of finding Particular Integrals :-

Method #1:

When $F(x)$ is an exponential function :-

$$y_p = \frac{1}{f(0)} e^{ax} = \frac{1}{f(a)} e^{ax}, \text{ if } f(a) = 0 \text{ then}$$

$$y_p = x \frac{1}{f'(a)} e^{ax} \text{ and again } f'(a) = 0 \text{ then}$$

$$y_p = x^2 \frac{1}{f''(a)} e^{ax} \text{ so on.}$$

Ex no 1

$$(i) \quad y''' + 3y'' + 3y' + y = e^{-x}$$

$$D^3y + 3D^2y + 3Dy + y = e^{-x}$$

$$(D^3 + 3D^2 + 3D + 1)y = e^{-x}$$

The characteristic equation is

$$D^3 + 3D^2 + 3D + 1 = 0$$

$$D = -1, -1, -1$$

The complementary solution

$$y_c = (C_1 + C_2x + C_3x^2)e^{-x}$$

Now For Particular Integral ν

$$y_p = \frac{1}{f(D)} e^{ax}$$

$$y_p = \frac{1}{(D^3 + 3D^2 + 3D + 1)} e^{-x}$$

$$\text{Put } D = -1$$

$$y_p = \frac{1}{(-1)^3 + 3(-1)^2 + 3(-1) + 1} =$$

$\rightarrow (0)$ undefined

Now take the derivative of $f(D)$

$$f'(D) = 3D^2 + 6D + 3$$

$$y_p = x \frac{1}{3D^2 + 6D + 3} e^{-x} = x \frac{1}{3(-1)^2 + 6(-1) + 3} e^{-x} \rightarrow (0)$$

Now again take derivative

$$y_p = x^2 \frac{1}{6D + 6} e^{-x} = x^2 \frac{1}{6(-1) + 6} e^{-x} \rightarrow (0)$$

Now again take derivative:

$$y_p = x^3 \frac{1}{6} e^{-x} = x^3 \frac{1}{6} e^{-x}$$

The General Solution is

$$y = y_c + y_p$$

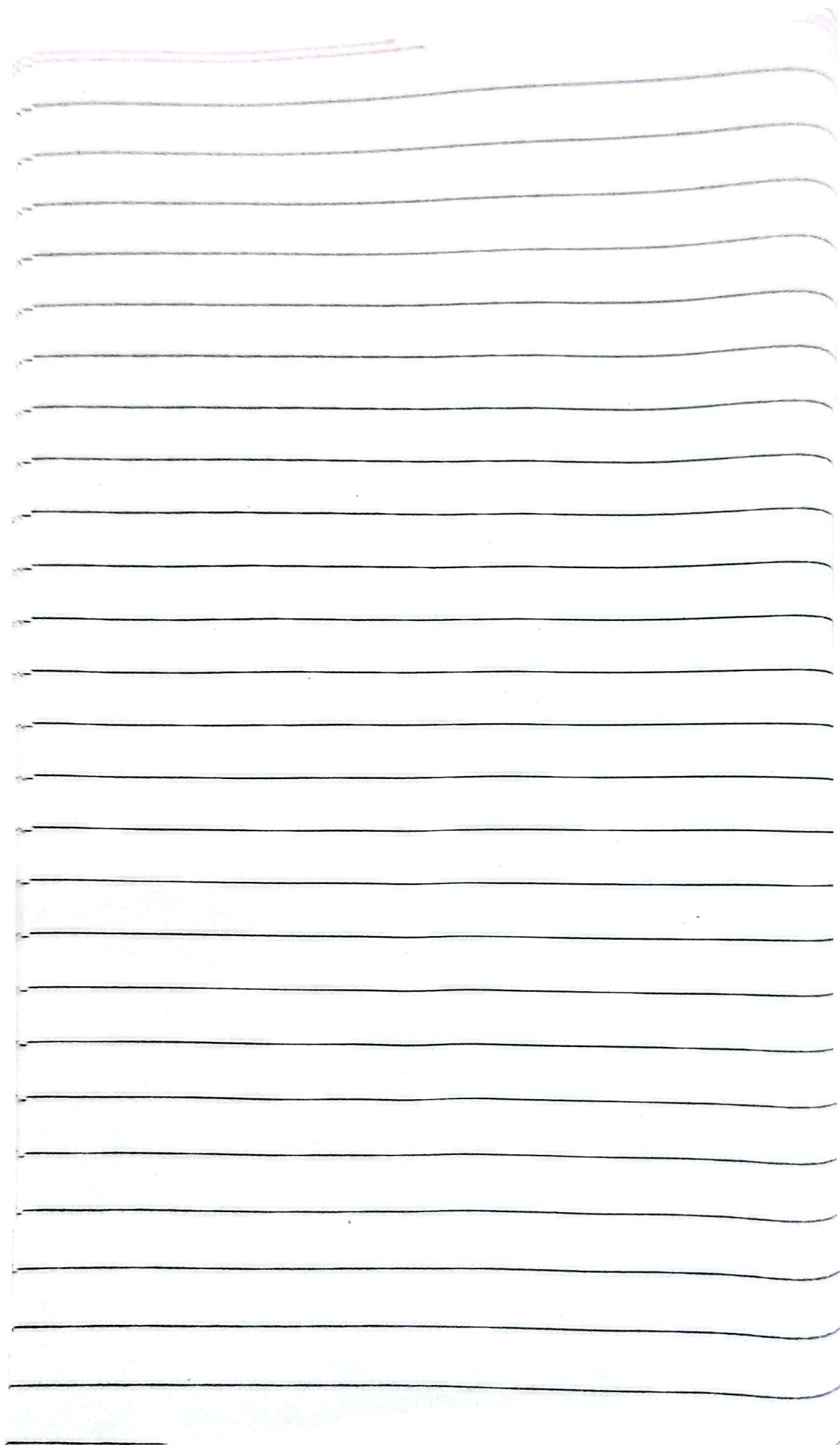
$$y = (C_1 + C_2 x + C_3 x^2) e^{-x} + x^3 \frac{1}{6} e^{-x}$$

Ans

Q $y'' - 6y' + 9y = 6e^{3x} + 7e^{-2x} - \ln 2$

Integral

$$y_p = \frac{1}{D^2 - 6D}$$



Method # 02

When $F(x)$ is an algebraic function

$$\frac{1}{f(D)} x^n = [f(D)]^{-1} \cdot x^n$$

Expand $[f(D)]^{-1}$ by using binomial theorem and proceed in ascending powers of D as the derivatives of the algebraic terms become zero.

The given below formula will be used

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - x^5 + \dots$$

Eg # 1

$$y''' + 4y = x^4$$

$$D^4 y + 4y = x^4$$

$$\Rightarrow (D^4 + 4)y = x^4$$

The characteristic eq. is

$$D$$

For y_p

$$y_p = \frac{1}{D^4+4} x^4$$
$$= \frac{1}{4\left(\frac{D^4}{4}+1\right)} x^4 \Rightarrow \frac{1}{4} \left[1 + \frac{D^4}{4}\right]^{-1} x^4$$

Expanding

$$y_p = \frac{1}{4} \left[1 \ominus \frac{D^4}{4} + \left(\frac{D^4}{4}\right)^2 \ominus \left(\frac{D^4}{4}\right)^3 + \dots \right] x^4$$

$$y_p = \frac{1}{4} \left[x^4 - \frac{D^4(x^4)}{4} + \frac{D^8(x^4)}{16} - \frac{D^{12}(x^4)}{64} + \dots \right]$$

$$D(x^4) = 4x^3 ; D^2(x^4) = 12x^2 ; D^3(x^4) = 24x ; D^4(x^4) = 24 ; D^5(x^4) = 0$$

$$y_p = \frac{1}{4} \left[x^4 - \frac{24}{4} + \frac{0}{16} - \frac{0}{64} + 0 \right]$$

$$y_p = \frac{x^4}{4} - \frac{3}{2}$$

The General Solution is

$$y = e^{-x} (C_1 \cos x + C_2 \sin x) + e^x (C_3 \cos x + C_4 \sin x) + \frac{x^4}{4} - \frac{3}{2}$$

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$$Q \quad D^2(D^2+4)y = 96x^2$$

$$D(x^2) = 2x$$

For y_p :

$$D^2(x^2) = 2$$

$$D^3(x^2) = 0$$

$$y_p = \frac{96x^2}{(D^4+4D^2)}$$

$$y_p = \frac{1}{(D^4+4D^2)} 96x^2$$

$$y_p = \frac{1}{4D^2 \left(\frac{D^4}{4D^2} + 1 \right)} 96x^2$$

$$y_p = \frac{1}{4D^2 \left(1 + \frac{D^2}{4} \right)} 96x^2$$

$$y_p = \frac{1}{4D^2} \left(1 + \frac{D^2}{4} \right)^{-1} \cdot \overset{24}{\cancel{96}} x^2$$

$$y_p = \frac{24}{D^2} \left(1 + \frac{D^2}{4} \right)^{-1} x^2$$

$$= \frac{24}{D^2} \left[1 - \frac{D^2}{4} + \left(\frac{D^2}{4} \right)^2 + \dots \right] x^2$$

$$= \frac{24}{D^2} \left[x^2 - \frac{D^2}{4} (x^2) + \frac{D^4}{16} (x^2) + \dots \right]$$

$$= \frac{24}{D^2} \left[x^2 - \frac{2}{4} + 0 + 0 \right]$$

$$= \frac{24}{D^2} \left[x^2 - \frac{1}{2} \right]$$

$$= \cancel{24} x^2 - 12$$

$$D^4(u + v) = 10x$$

$$D(x^2) = 2x$$

For y_p :

$$D^2(x^2) = 2$$

$$D^3(x^2) = 0$$

$$y_p = \frac{96x^2}{(D^4 + 4D^2)}$$

$$y_p = \frac{1}{(D^4 + 4D^2)} 96x^2$$

$$y_p = \frac{1}{4D^2 \left(\frac{D^4}{4D^2} + 1 \right)} 96x^2$$

$$y_p = \frac{1}{4D^2 \left(1 + \frac{D^2}{4} \right)} 96x^2$$

$$y_p = \frac{1}{4D^2} \left(1 + \frac{D^2}{4} \right)^{-1} \cdot \frac{24}{48} 96x^2$$

$$y_p = \frac{24}{D^2} \left(1 + \frac{D^2}{4} \right)^{-1} x^2$$

$$= \frac{24}{D^2} \left[1 - \frac{D^2}{4} + \left(\frac{D^2}{4} \right)^2 + \dots \right] x^2$$

$$= \frac{24}{D^2} \left[x^2 - \frac{D^2(x^2)}{4} + \frac{D^4(x^2)}{16} + \dots \right]$$

$$= \frac{24}{D^2} \left[x^2 - \frac{2}{4} + 0 + 0 \right]$$

$$= \frac{24}{D^2} \left[x^2 - \frac{1}{2} \right]$$

$$= \frac{24x^2 - 12}{D^2}$$

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$$y_p = \frac{24}{D^3} \left[x^3 - \frac{1}{2} \right]$$

$$y_p = \frac{24}{D} \left[\frac{x^3}{3} - \frac{1}{2} \right] \quad \text{Integrating}$$

$$y_p = 24 \left[\frac{x^4}{12} - \frac{x^2}{4} \right]$$

$$y_p = \frac{24}{4} \left[\frac{x^4}{3} - x^2 \right]$$

$$y_p = 6 \left[\frac{x^4 - 3x^2}{3} \right]$$

$$\boxed{y_p = 2x^4 - 6x^2}$$

The Complementary Equation

$$\text{Ex 3 } (D^2 + 5D + 4)y = 3 - 2x$$

$$y_p = \frac{1}{(D^2 + 5D + 4)} \cdot (3 - 2x)$$

$$y_p = \frac{1}{4 \left(\frac{D^2 + 5D + 1}{4} \right)} \cdot (3 - 2x)$$

$$y_p = \frac{1}{4} \left\{ 1 + \left(\frac{D^2 + 5D}{4} \right) \right\}^{-1} \cdot (3 - 2x)$$

$$y_p = \frac{1}{4} \left\{ 1 - \left(\frac{D^2 + 5D}{4} \right) + \left(\frac{D^2 + 5D}{4} \right)^2 + \dots \right\} \cdot (3 - 2x)$$

$$y_p = \frac{1}{4} \left\{ 1 - \left(\frac{D^2 + 5D}{4} \right) + \left(\frac{D^4 + 10D^3 + 25D^2}{16} \right) + \dots \right\} \cdot (3 - 2x)$$

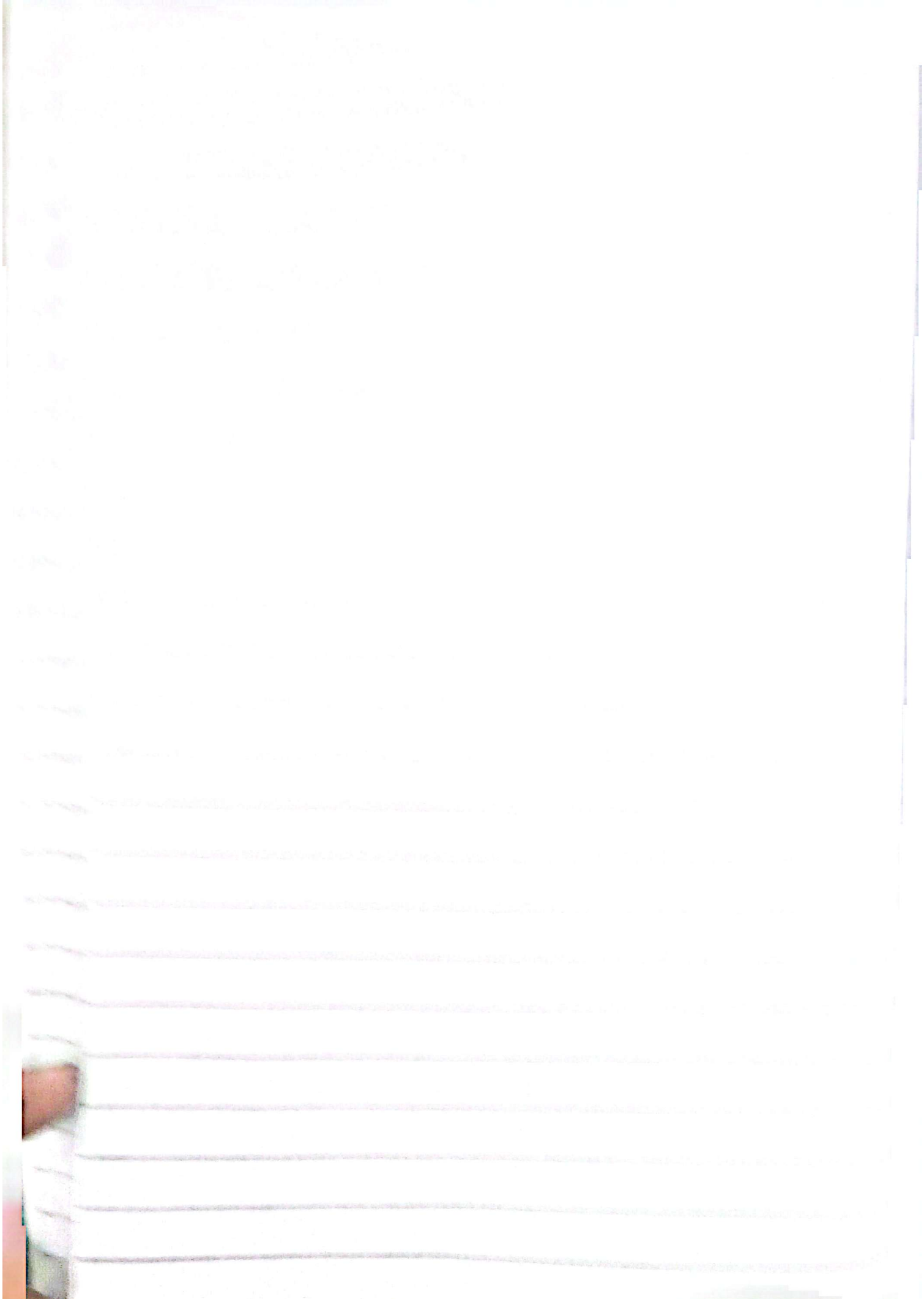
$$y_p = \frac{1}{4} \left\{ (3 - 2x) - \frac{(D^2 + 5D)(3 - 2x)}{4} + \frac{D^4(3 - 2x) + 10D^3(3 - 2x)}{16} - \frac{25D^2(3 - 2x) + \dots}{4} \right\}$$

$$y_p = \frac{1}{4} \left\{ (3 - 2x) + \frac{(-0) + 5(-2)}{4} + \frac{0 + 10(0) + 25(0) + 0}{16} \right\}$$

$$y_p = \frac{1}{4} \left\{ (3 - 2x) + \frac{10}{4} + 0 \right\}$$

$$y_p = \frac{1}{4} \left\{ \frac{6 - 4x + 5}{2} \right\}$$

$$y_p = \frac{11 - 4x}{8}$$



Method # 3

When $F(x)$ is the function of Sine or Cosine
 : Applicable only of sine and cosine function and
 also applicable only for D^2 , not for $D, D^3, D^4, \dots$

$$\frac{1}{f(D^2)} \sin ax = \frac{1}{f(-a^2)} \sin ax$$

$$\text{or } \frac{1}{f(D^2)} \cos ax = \frac{1}{f(-a^2)} \cos ax$$

If $f(-a^2) = 0$ then

$$\frac{1}{f(D^2)} \sin ax = x \frac{1}{f'(D^2)} \sin ax.$$

$$\text{or } \frac{1}{f(D^2)} \cos ax = x \frac{1}{f'(D^2)} \cos ax.$$

and proceed.

$$\text{Ex 1} \quad (D^2 + 4)y = \sin 3x$$

For y_p :

$$y_p = \frac{1}{D^2 + 4} \sin 3x$$

$$y_p = \frac{1}{-3^2 + 4} \sin 3x = \boxed{-\frac{1}{5} \sin 3x}$$

Ex no 2# $(D^2+4)y = \cos 2x$

Find y_c in h.w

$$y_p = \frac{1}{(D^2+4)} \cos 2x$$

$$y_p = \frac{1}{-4+4} \cos 2x$$

↓
0

$$y_p = x \frac{1}{(2D)} \cos 2x$$

$$y_p = x \frac{1}{2(-4)} \cos 2x$$

$$y_p = -\frac{x}{8} \cos 2x$$

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$$y_p = \frac{x}{20} \cos 2x$$

Taking Integrals

$$y_p = \frac{x}{2} \left(\frac{\sin 2x}{2} \right) = \boxed{\frac{x \sin 2x}{4}}$$

Ex No: 3

$$(D^2 + D + 1)y = \cos 2x$$

$$y_p = \frac{1}{(D^2 + D + 1)} \cos 2x$$

Put $D = -2$ only in square

$$y_p = \frac{1}{-(+2)^2 + D + 1} \cos 2x = \frac{1}{D - 3} \cos 2x$$

Rationalizing

$$y_p = \frac{1}{(D - 3)} \times \frac{(D + 3)}{(D + 3)} \cos 2x$$

$$y_p = \frac{D + 3}{D^2 - 9} (\cos 2x)$$

$$\text{Put } D^2 = -(2)^2$$

$$y_p = \frac{D + 3}{-4 - 9} \cos 2x$$

$$y_p = \frac{D + 3}{-13} \cos 2x$$

Now taking derivative

$$y_p = \frac{-1}{13} \{ D(\cos 2x) + 3\cos 2x \}$$

$$y_p = \frac{-1}{13} \{ -2\sin 2x + 3\cos 2x \}$$

$$y_p = \frac{2\sin 2x - 3\cos 2x}{13}$$

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The y_c and also General Solution

Method No 4 (Shifting Method)

In case of product of two terms in which one is an exponential and other is any function (algebraic, trigonometric etc). We can take exponential before integral if and only if we add the coefficient of exponential that is e^{ax} with D then use any method to solve regarding the function.

Ex no 1

$$(D^2 + 2D + 4)y = e^x \cdot \sin 2x$$

For y_p

$$y_p = \frac{1}{(D^2 + 2D + 4)} e^x \cdot \sin 2x$$

$$y_p = e^x \cdot \frac{1}{(D+1)^2 + 2(D+1) + 4} \sin 2x$$

$$\frac{1}{D^2 + 2D + 1 + 2D + 2 + 4} \sin 2x$$

$$y_p = e^x \cdot \frac{1}{D^2 + 4D + 7} \sin 2x$$

Using method 3

$$y_p = e^x \cdot \frac{1}{-(-2)^2 + 4D + 7} \sin 2x$$

$$y_p = e^x \cdot \frac{1}{-4 + 4D + 7} \sin 2x$$

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$$y_p = e^x \cdot \frac{1}{4D+3} \sin 2x$$

$$y_p = e^x \cdot \frac{1}{(4D+3)} \times \frac{(4D-3)}{(4D-3)} \sin 2x$$

$$y_p = e^x \cdot \frac{4D-3}{16D^2-9} \sin 2x$$

$$y_p = e^x \cdot \frac{4D-3}{-16(2)^2-9} \sin 2x$$

$$y_p = e^x \cdot \frac{4D-3}{-64-9} \sin 2x$$

$$y_p = e^x \cdot \frac{4D-3}{-73} \sin 2x$$

$$y_p = (e^x) \left(\frac{-1}{73} \right) \{ 0 \cos 2x - 3 \sin 2x \}$$

Higher Order Ordinary Differential Equations With Variable Coefficients (Cauchy's Differential Eq)

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_{n-2} x^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots$$

$$a_0 y = \phi(x) \rightarrow (1)$$

here $a_0, a_1, a_2, \dots, a_n$ are constants equation (1) is homogenous if $\phi(x) = 0$. Otherwise, non-homogenous

Put

$$x = e^t, \quad t = \ln x$$

$$\frac{d}{dx} = D$$

$$\& \quad xD = \Delta$$

$$x^2 D^2 = \Delta(\Delta-1)$$

Example No: 1

$$x^2 \frac{d^2 y}{dx^2} + 7x \frac{dy}{dx} + 5y = x^5$$

$$x^2 D^2 y + 7x Dy + 5y = x^5$$

$$y (x^2 D^2 + 7xD + 5)y = x^5 \rightarrow (1)$$

Let $x = e^t$ and $t = \ln(x)$

$$xD = \Delta \quad \text{and} \quad x^2 D^2 = \Delta(\Delta-1) = \Delta^2 - \Delta$$

Substitute them in eq (1)

$$x' = e^t$$

$$x^2 = e^{2t} \dots$$

$$\{(\Delta^2 - \Delta) + 7\Delta + 5\} y = e^{5t}$$

The characteristic eq is:-

$$\Delta^2 + 6\Delta + 5 = 0$$

$$\Delta^2 + 5\Delta + \Delta + 5 = 0$$

$$\Delta(\Delta+5) + 1(\Delta+5) = 0$$

$$\Delta = -1 \quad \text{and} \quad \Delta = -5$$

The complementary function is:-

$$y_c = C_1 e^{-t} + C_2 e^{+5t} \rightarrow (A)$$

For y_p

$$y_p = \frac{1}{(\Delta^2 + 6\Delta + 5)} \cdot e^{+5t}$$

$$y_p = \frac{1}{(+5)^2 + 6(+5) + 5} e^{+5t} = \frac{1}{25 + 30 + 5} e^{5t} = \frac{1}{60} e^{5t}$$

The General Solution

$$y = y_c + y_p$$

$$y = C_1 e^{-t} + C_2 e^{-5t} + \frac{1}{60} e^{5t}$$

$$\text{Put } t = \ln x$$

$$y = C_1 e^{-\ln x} + C_2 e^{-5 \ln x} + \frac{1}{60} e^{5 \ln(x)}$$

$$y = C_1 \left(\frac{1}{x} \right) + C_2 \left(\frac{1}{x^5} \right) + \frac{1}{60} x^5$$

$$y = C_1 x^{-1} + C_2 x^{-5} + \frac{1}{60} x^5$$

Example No: 2

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 5y = x^2 \sin(\ln x)$$

$$x^2 D^2 y - 3x Dy + 5y = x^2 \sin(\ln(x))$$

$$(x^2 D^2 - 3xD + 5)y = x^2 \sin(\ln x)$$

$$x^2 D^2 = \Delta(\Delta-1) \quad ; \quad xD = \Delta$$

$$\{(\Delta^2 - \Delta) - 3\Delta + 5\}y = e^{2t} \sin t$$

$$(\Delta^2 - \Delta - 3\Delta + 5)y = e^{2t} \sin t$$

$$(\Delta^2 - 4\Delta + 5)y = e^{2t} \sin t$$

The characteristic eq is

$$(\Delta^2 - 4\Delta + 5) = 0$$

$$y = C_1 x^2 \cos(\ln x) + C_2 x^2 \sin(\ln x) - \frac{1}{2} x^2 (\ln x) \cos(\ln x)$$

Date: _____

By using Quadratic eq

$$\Delta = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(5)}}{2(1)}$$

$$\Delta = \frac{4 \pm \sqrt{16 - 20}}{2} = \frac{4 \pm \sqrt{-4}}{2} = \frac{4 \pm 2i}{2} = \boxed{2 \pm i}$$

$$\Delta = 2 + i \quad \text{and} \quad \Delta = 2 - i$$